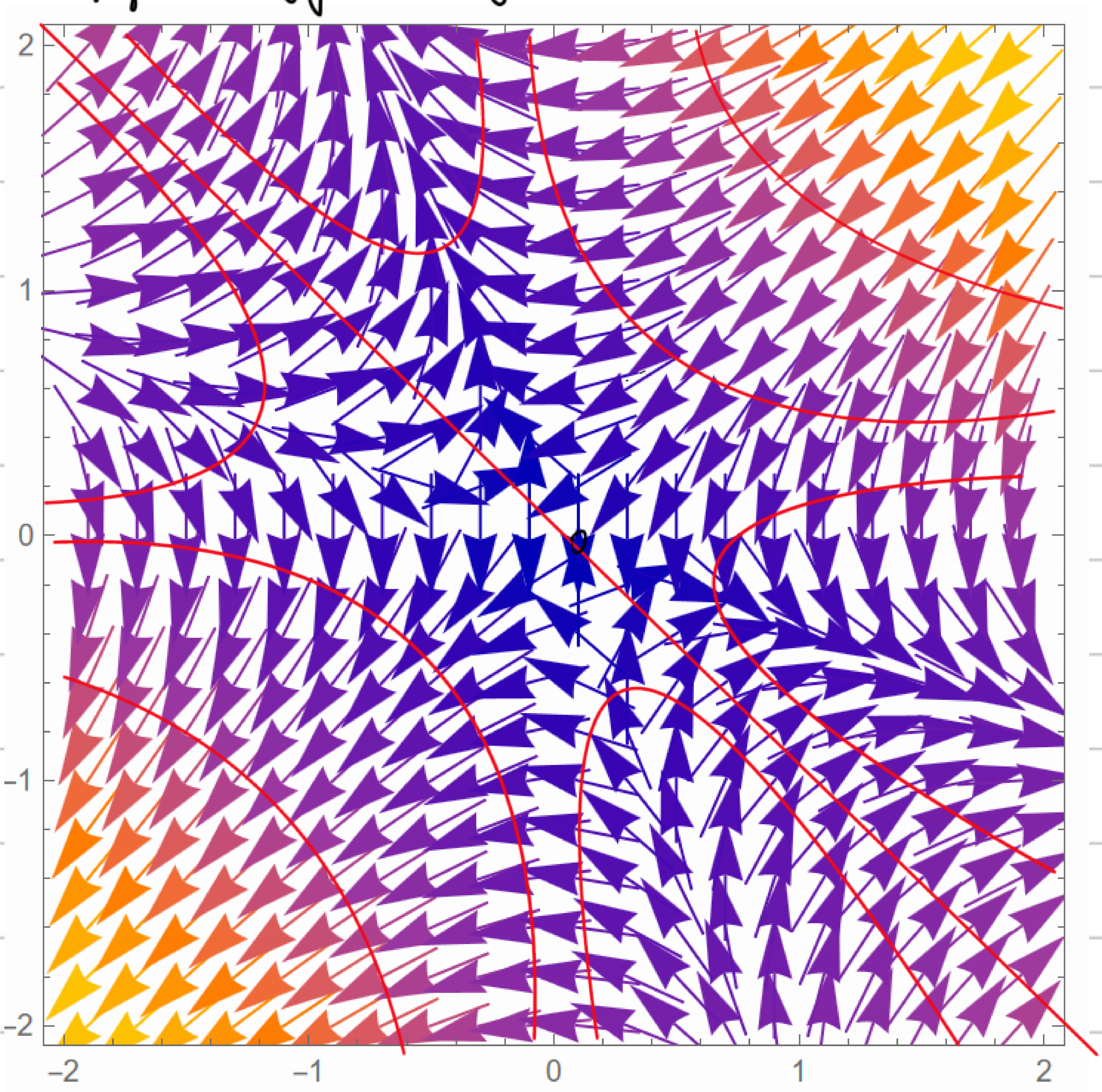
Guojiao Zhao 523370910264.

P1. (a) $F_x = -\frac{\partial u}{\partial x} = -y^2 - 2yx$

$F_y = -\frac{\partial u}{\partial y} = -2xy - x^2.$

(a)(b)



(c) the graph shows that the force field is perpendicular to the equipotential lines and points towards regions of lower potential

(d) $y=x$

$W_1 = \int_L F dr = \int_0^1 (-3x^2)\,dx + \int_0^1 (-3x^2)\,dx$

$= -2\ J$

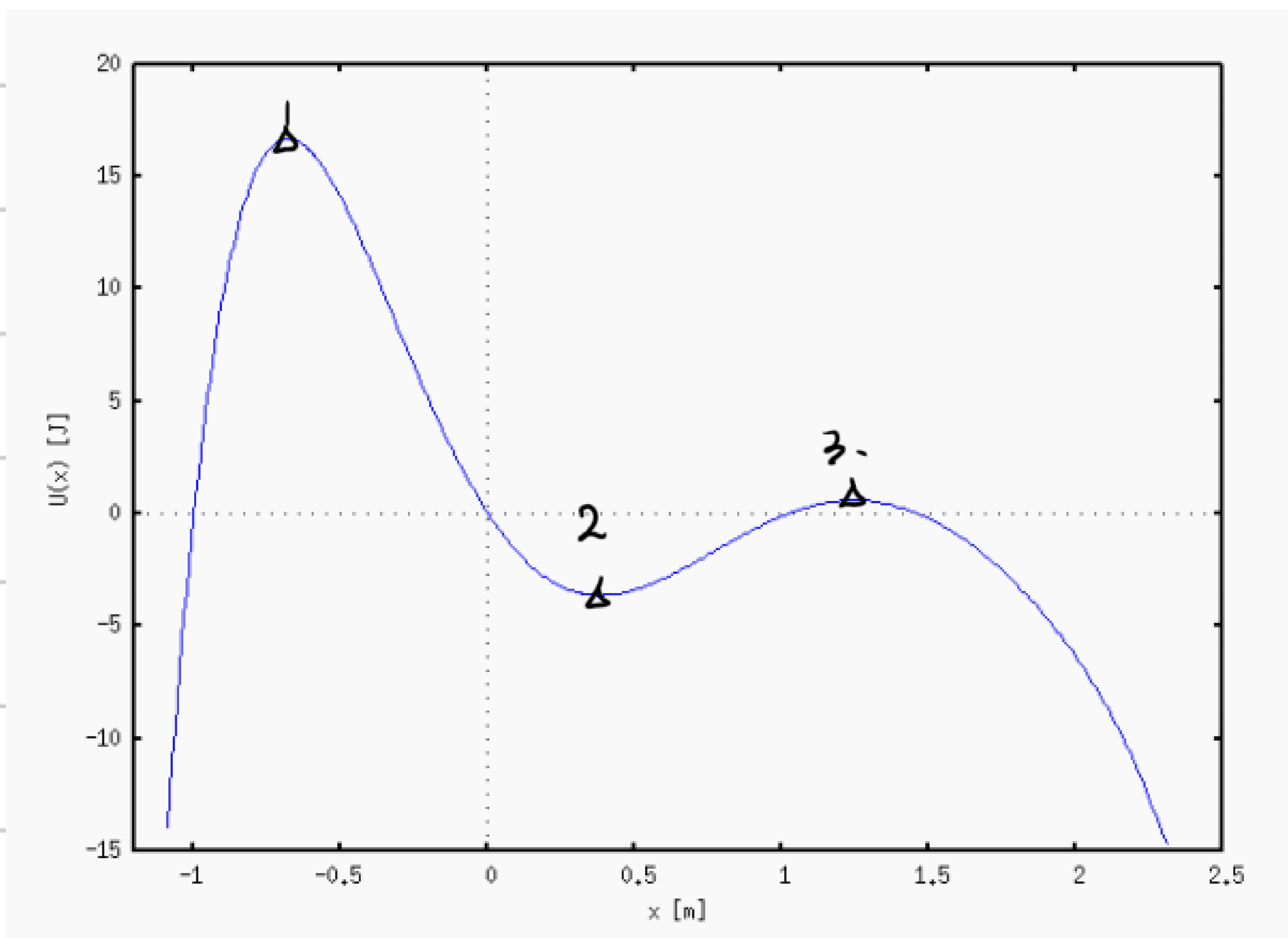
$y = x^2 \qquad \bar{F} = (-x^4 - 2x^3,\ -2x^3 - x^2) \qquad \frac{dy}{dx} = 2x$

$W_2 = \int_{parabola} F dx = \int_0^1 (-x^4 - 2x^3)\,dx + \int_0^1 (-2x^3 - x^2)\,2x\,dx.$

$= -\frac{1}{5} - \frac{1}{2} - \frac{4}{5} - \frac{1}{2}$

$= -2\ J$

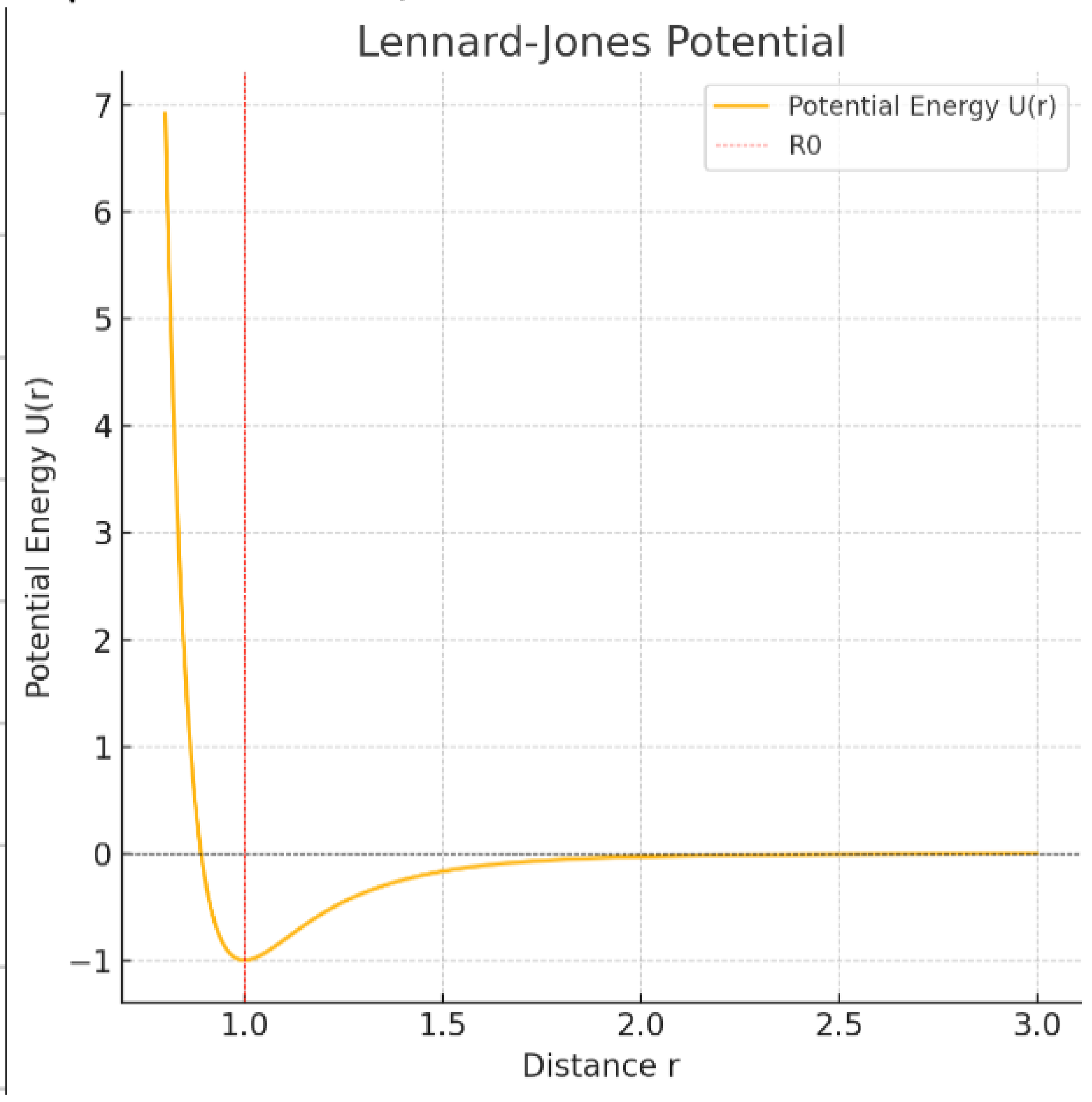
P2.

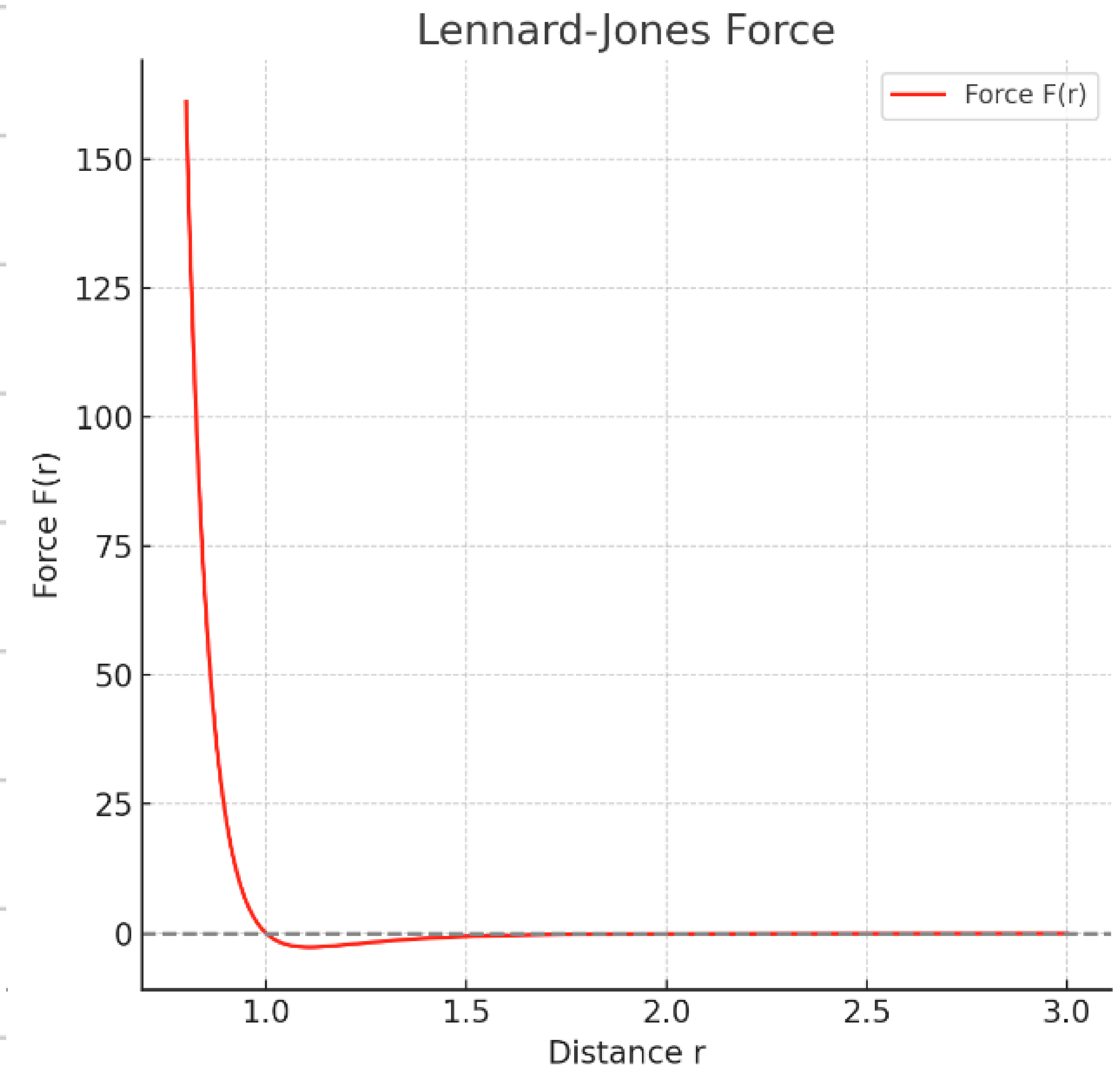


All the equilibrium points, marked in graph, are the points when $\frac{dU(x)}{dx}=0$

points 1.3 are unstable ; point 2 is stable.

P3. (a) $F=-\frac{du}{dr} = +12U_0R_0^{12}r^{-13} - 12U_0R_0^{6}r^{-7}$





Repulsion: $+12U_0R_0^{12}r^{-13}$ attraction: $-12U_0R_0^{6}r^{-7}$

(b) when $r=R_0$ $F=0$, the potential energy is the lowest, to be $-U_0$

(c) $x=r-R_0$

$$F=+12U_0R_0^{12}(x+R_0)^{-13} - 12U_0R_0^{6}(x+R_0)^{-7}$$

$$U(x) = U_0\left[\left(\frac{R_0}{x+R_0}\right)^{12} - 2\left(\frac{R_0}{x+R_0}\right)^{6}\right]$$

$$\approx U(0) + \frac{1}{2!}U''(0)x^2 = -U_0 + 36U_0R_0^{-2}x^2$$

$$F(x) = -\frac{dU(x)}{dx} = -72U_0R_0^{-2}x$$

$K = 72U_0R_0^{-2}$ $\omega_0 = \sqrt{\frac{k}{m}} = \sqrt{\frac{72U_0R_0^{-2}}{m}}$

$$T = \frac{2\pi}{\omega_0} = 2\pi\sqrt{\frac{m}{72U_0R_0^{-2}}} = \frac{\pi R_0}{3}\sqrt{\frac{m}{2U_0}}$$

the atoms or molecules oscilliate around $r=R_0$ within a small range of displacement $\left|\frac{x}{R_0}\right| \ll 1$

P4. $F(x) = -\frac{dU}{dx} = 2\alpha U_0 \tan\alpha x \frac{1}{\cos^2\alpha x}$ $U''(x) = +2\alpha^2U_0\frac{1+3\sin^2\alpha x}{\cos^4\alpha x}$

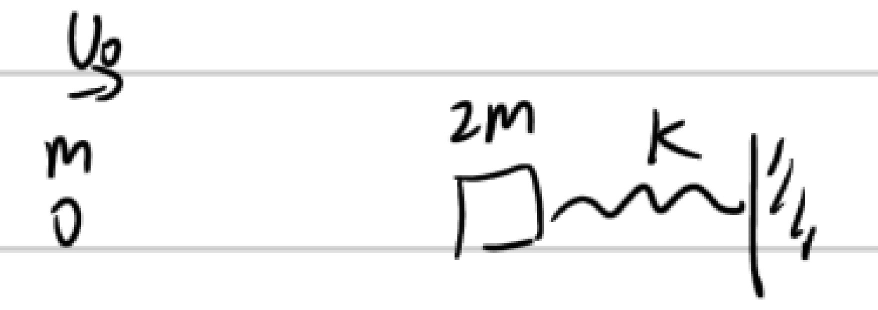
when $F(x)=0$ $\alpha x_0 = n\pi$ $(n\in\mathbb{Z})$ $x_0 = \frac{n\pi}{\alpha}$ $\therefore 0, \frac{\pi}{\alpha}, \frac{2\pi}{\alpha} \ldots$

$U(x) = U(x_0) + \frac{1}{2!}U''(x_0)(x-x_0)^2$ $U''(x_0) = +2\alpha^2U_0$

$u(x) \approx -\alpha^2 U_0 (x-x_0)^2$ $\qquad k = U''(x_0) = -2\alpha^2 U_0$

$\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{2\alpha^2 U_0}{m}}$ $\qquad T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{m}{2\alpha^2 U_0}} = \frac{\pi}{\alpha}\sqrt{\frac{2m}{U_0}}$

P5.

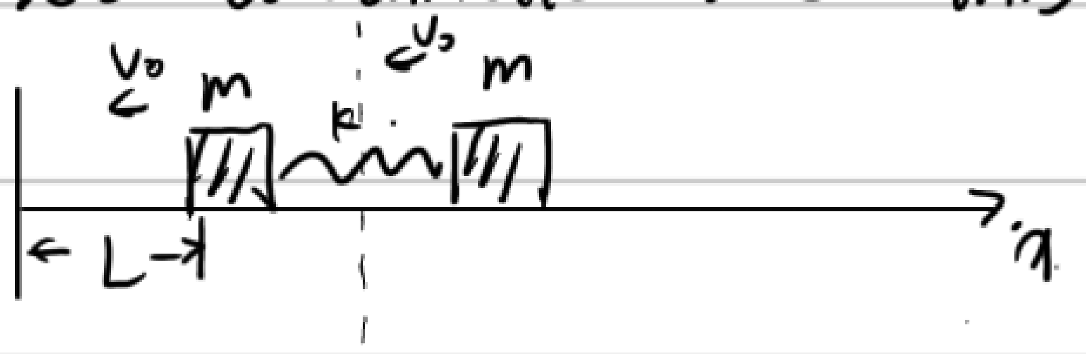


take the Earth as FoR.

$$\begin{cases} mv_0 = mv_1 + 2mv_2 \\ \frac12 mv_0^2 = \frac12 mv_1^2 + \frac12\cdot 2mv_2^2 \end{cases} \Rightarrow \begin{aligned} v_1 &= \frac{m-2m}{m+2m}v_0 = -\frac13 v_0 \\ v_2 &= \frac{2m}{m+2m}v_0 = \frac23 v_0 \end{aligned}$$

$\frac12 \times 2m\, v_2^2 = \frac12 k x^2$ $\qquad x = \sqrt{\frac{8m}{9k}}\, v_0$

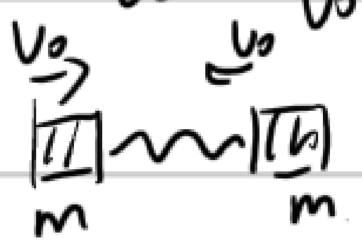
P6. set coordinate like this



$\bar r_{cm} = \frac{m r_1 + m r_2}{2m}$

Before collision : $t_0 = \frac{L}{v_0}$

After collision :



$\omega = \sqrt{\frac{2k}{m}}$

until the spring return to original length . $\quad t_1 = \frac12 T = \frac12 \frac{2\pi}{\omega} = \pi\sqrt{\frac{m}{2k}}$

at this point, collide again

$t_2 = t_0 = \frac{L}{v_0}$

so the total time $t_{total} = t_0 + t_1 + t_2 = \frac{2L}{v_0} + \pi\sqrt{\frac{m}{2k}}$

P7 $\quad M\bar v_{cm} = \int_0^t \bar F\, dt$ $\qquad M\frac{d\bar r_{cm}}{dt} = (0.05t, 0, 0)$

$\int_{\bar r_{c0}}^{\bar r} M\, d\bar r_{cm} = \left(\int_0^2 0.05t\, dt, 0, 0\right)$

$M(\bar r - \bar r_{c0}) = (0.1, 0, 0)$ $\qquad \bar r - \bar r_{c0} = (\frac{10}{3} m, 0, 0) = (\frac13 \times 10^3 cm, 0, 0)$

since $\quad \bar r_{c0} = \frac{m_1\bar r_1 + m_2\bar r_2 + m_3\bar r_3}{m_1+m_2+m_3} = (\frac76, 2, \frac{17}{6})$ cm

so $\bar r = (\frac{669}{2}, 2, \frac{17}{6})$ cm

P8.

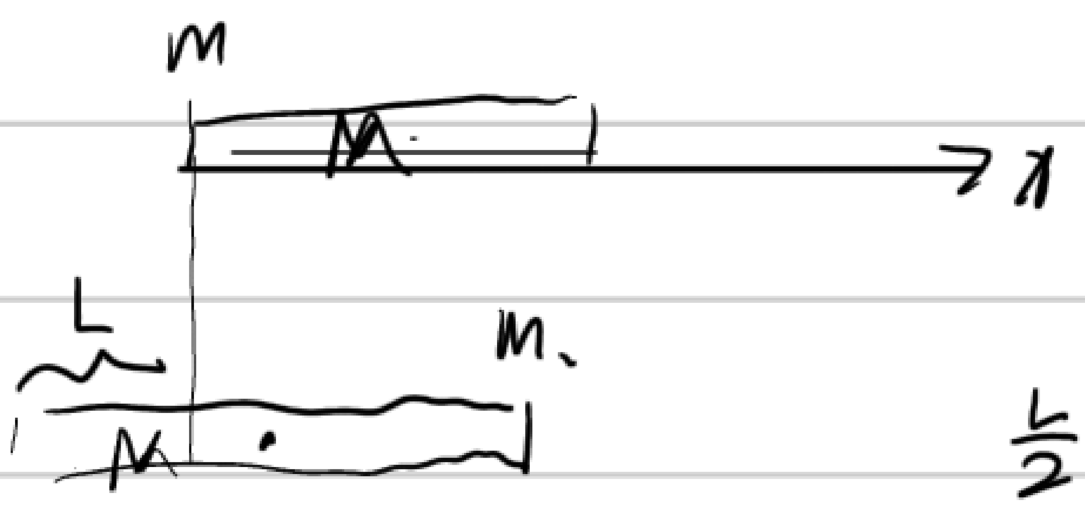
$dm = dx\cdot A\cdot\frac{m}{Al} = dx\frac{m}{l}$.

(a) $x_{cm} = \frac{\int x\, dm}{\int dm} = \frac{\int_0^l \frac{m}{l} x\, dx}{m} = \frac{\frac{m}{2l} l^2}{m} = \frac12 l$

(b) $x_{cm} = \frac{\int x\, dm}{\int dm} = \frac{\int_0^l A\alpha x^2\, dx}{\int A\alpha x\, dx} = \frac{\frac13 A\alpha l^3}{\frac12 A\alpha l^2} = \frac23 l.$

$dm = dx\, A\cdot\alpha x.$

P9.



$$x_{cm} = \frac{0 + \frac{L}{2}M}{m+M}$$

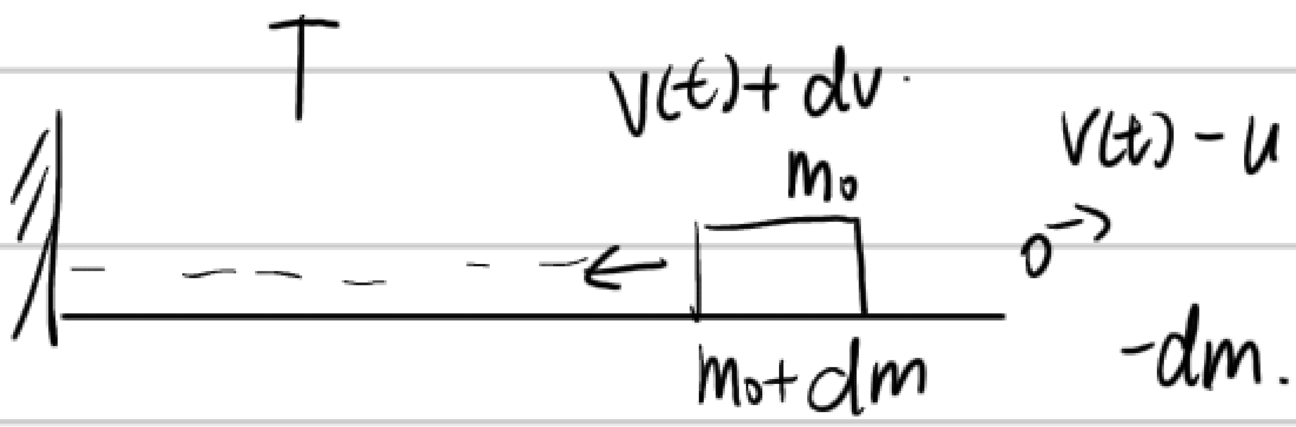
$$x_{cm} = \frac{(L-d)m + (\frac{L}{2} - d)M}{m+M}$$

$$\frac{L}{2}\frac{M}{M+m} = \frac{(L-d)m + (\frac{L}{2} - d)M}{M+m}$$

$$dM = (L-d)m$$

$$d = \frac{m}{M+m}L$$

P10.



$$M(t)V(t) = (M(t)+dm)(V(t)+dv) - dm(V(t)-u)$$

$$= M(t)V(t) + M(t)dv + dm V(t) + dm dv - dm V(t) + dm u$$

$$M(t)\frac{dv}{dt} = -u\frac{dm}{dt}$$

$$a = -\frac{u}{M(t)}\frac{dm}{dt} = \frac{u}{M(t)}\alpha = \frac{u}{m_0 - \alpha t}\alpha.$$

$$V(t) = \int_0^T a\,dt = \int_0^T \frac{u\alpha}{m_0 - \alpha t}dt = u\ln\frac{m_0}{m_0 - \alpha T}$$

notice that $a(t)$ keeps the same

$$-V(T) + \int_T^{T+\Delta T_1} a\,dt = 0 \qquad u\ln\frac{m_0 - T\alpha}{m_0 - T\alpha - t\alpha} = u\ln\frac{m_0}{m_0 - \alpha T}$$

$$m_0(m_0 - T\alpha - t\alpha) = (m_0 - \alpha T)(m_0 - T\alpha)$$

$$\cancel{m_0^2} - m_0T\alpha - t\alpha m_0 = \cancel{m_0^2} - 2m_0T\alpha + \alpha^2T^2$$

$$-t\alpha m_0 = -m_0T\alpha + \alpha^2T^2$$

$$t = \frac{-m_0T\alpha + \alpha^2T^2}{-m_0\alpha} = T - \frac{\alpha T^2}{m_0}$$

$$t = 10 - \frac{0.01 \times 10^2}{1} = 9s$$